Write-up

An analyst co-pilot that clarifies, checks definitions, picks the right data layer and shows its evidence.

01 What I built and why

1 Clarify — metric, period, population; asks if ambiguous

2 Verify definition — metric exists and its definition in registry + catalog

3 Route to a layer — Gold if it fits; Silver for detail; Bronze for raw audit

4 Query + evidence — read-only, bounded; one log per query, never overwritten

5 Test hypotheses — ≤3 per round, ≤3 rounds; stop when explained

6 Report — pattern ≠ cause; fixed PDF template, ≤10 pages

Read by every run, in Claude Code or Codex: AGENTS.md · SKILL.md · catalog · registry

Guardrails: read-only authorizer · row and time limits

Gold — 8 monthly tables · fast standard answers

Silver — 10 cleaned tables · detail for investigations

Bronze — 20 views over raw tables · audits

One SQLite database; layers built in one transaction with 122 reconciliation checks. Step 3 decides which layer each query reads.

- **Clarifies instead of assuming.** When the metric, period or meaning is open, it asks or states a clear choice and records its assumptions.
- **Verifies the metric before computing it.** It confirms the metric exists and uses its approved definition and SQL from the metric registry, plus joins and known gaps from the data catalog. Unsupported requests are declined, not improvised.
- **Chooses the layer that fits.** Gold serves supported summaries; Silver preserves detail; Bronze provides source evidence. Correctness and coverage take priority over speed.
- **Evidence for every number and LLM-neutral by design.** Each query is logged with its SQL, layer and metric version. Definitions, rules and tools are shared across runtimes; identical behavior is not guaranteed. A reference website explains the workflow.

02 How it performed: 18 blind test questions, two rounds

	Q01	Q02	Q03	Q04	Q05	Q06	Q07	Q08	Q09	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18
Round 1	Pass	Fail	Pass	Pass	Pass	Pass	Fail	Pass	Pass	Pass	Partial	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Round 2	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Partial	Pass	Pass	Pass	Pass	Pass	Pass	Pass

Round 1: 15 pass · 1 partial · 2 fail **Round 2: 17 · 1 · 0**

Question type	Questions	Result	What happened
Simple numbers and filters	Q01 Q05 Q08 Q17 Q18	All pass	Exact to the user and the cent
Vague questions	Q03 Q10 Q15 Q18	All pass	Asked, or stated a clear choice, before picking a metric, period or meaning
Unanswerable question	Q16	Pass	Declined and named the data that's missing
Data-quality traps	Q12 Q13 Q14	All pass	Caught both planted issues; didn't claim to have detected a reviewed one
Comparisons and lifecycle	Q04 Q06 Q09	All pass	Right time basis; groups reconcile to the total
Cohort conversion	Q07	Fail → Pass	Treated April, the start-up month, as a normal baseline; a new skill rule fixed it
Funnel drop-off	Q11	Partial	Right answer, but read April's start-up as a funnel decline; the rule didn't trigger
GM investigation	Q02	Fail → Pass	v1 right on numbers but too firm on some conclusions; after three certainty rules, v2 and a blind rerun both passed

Scored against expected values computed from the raw tables, not the agent's layers. Correct numbers alone don't pass: a silent choice of definition, a missing caveat or an overstated conclusion fails. Round 2: Q02 was revised after review and rerun blind; Q07 and Q11 were rerun, with Q01 and Q09 to check nothing broke; other results carry over. **Hardest:** comparisons that need context the data doesn't announce, such as a start-up month.

03 Proposed rollout for a 20-person product team

Time to answer (observed) | **Simple** (MAU by month): 2–3 min | **Normal** (refund impact): 5–6 min | **Complex** (MAU up, revenue flat): 10–12 min

- **How the team uses it:** analysts and PMs ask in plain English from Claude Code or Codex (later Slack or a web page), all on one shared repository, so definitions and guardrails are the same for every person and model.
- **Set-up:** build a data catalog and metric registry for the team's warehouse, with the analytics lead approving each definition, then the Gold and Silver layers the agent routes to.
- **Cost:** model usage and any warehouse costs; local SQLite has no query service fee. Not measured here: log tokens per question, then budget by question type. Answering from the smallest layer that fits keeps both cost and latency down.
- **Guardrails:** read-only access (database role plus the query tool's check), row and time limits, an evidence log behind every figure, no sensitive columns in the catalog, and a question back when the ask is unclear.
- **Freshness:** real data keeps arriving, unlike this dataset; provide a freshness tracker (last load per source and layer) so the agent can warn or stop when data is stale.
- **Quality over time:** a test set of questions with known answers (these 18, growing from real ones), rerun on every change to the skill, definitions or data; track pass rate, clarification rate, query errors and user ratings; turn failures into rules.

04 With more time

- Run the same questions in both runtimes for a direct Claude vs Codex comparison, plus held-out questions.
- Measure tokens and time per question to put a real cost on each question type.
- A helper that drafts the catalog and metric registry for a new database; a Slack or web front end.

Tools: Claude Code (Claude Opus 5.5, medium) built the framework, ran the GM investigation and the answer key; Codex (GPT Terra 6.1, medium) ran the evaluation; ChatGPT (Terra 5.6, later 6.1, medium) reviewed independently.